

```
import pandas as pd

df = pd.read_csv('Superstore.csv', encoding='latin1')
df.head()
```

	_Name	Segment	Country
	» Gute	Consumer	United States
	» Gute	Consumer	United States
	in Huff	Corporate	United States
	Donnel	Consumer	United States
	Donnel	Consumer	United States

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.columns
```

```
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
       'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
       'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
       'Product_Name', 'Sales'],
      dtype='object')
```

```
list(df.columns)
```

```
['Row_ID',
 'Order_ID',
 'Order_Date',
 'Ship_Date',
 'Ship_Mode',
 'Customer_ID',
 'Customer_Name',
 'Segment',
 'Country',
 'City',
 'State',
 'Postal_Code',
 'Region',
 'Product_ID',
 'Category',
 'Sub_Category',
 'Product_Name',
 'Sales']
```

```
df.columns = df.columns.str.strip()
list(df.columns)
```

```
['Row_ID',
 'Order_ID',
 'Order_Date',
 'Ship_Date',
 'Ship_Mode',
 'Customer_ID',
 'Customer_Name',
 'Segment',
 'Country',
 'City',
 'State',
 'Postal_Code',
 'Region',
 'Product_ID',
 'Category',
 'Sub_Category',
 'Product_Name',
 'Sales']
```

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
df.dtypes
```

	0
Row_ID	int64
Order_ID	object
Order_Date	datetime64[ns]
Ship_Date	object
Ship_Mode	object
Customer_ID	object
Customer_Name	object
Segment	object
Country	object
City	object
State	object
Postal_Code	float64
Region	object
Product_ID	object
Category	object
Sub_Category	object
Product_Name	object
Sales	float64

dtype: object

```
# Total Sales
total_sales = df['Sales'].sum()
```

```
# Total number of orders
```

```
total_orders = df['Order_ID'].nunique()

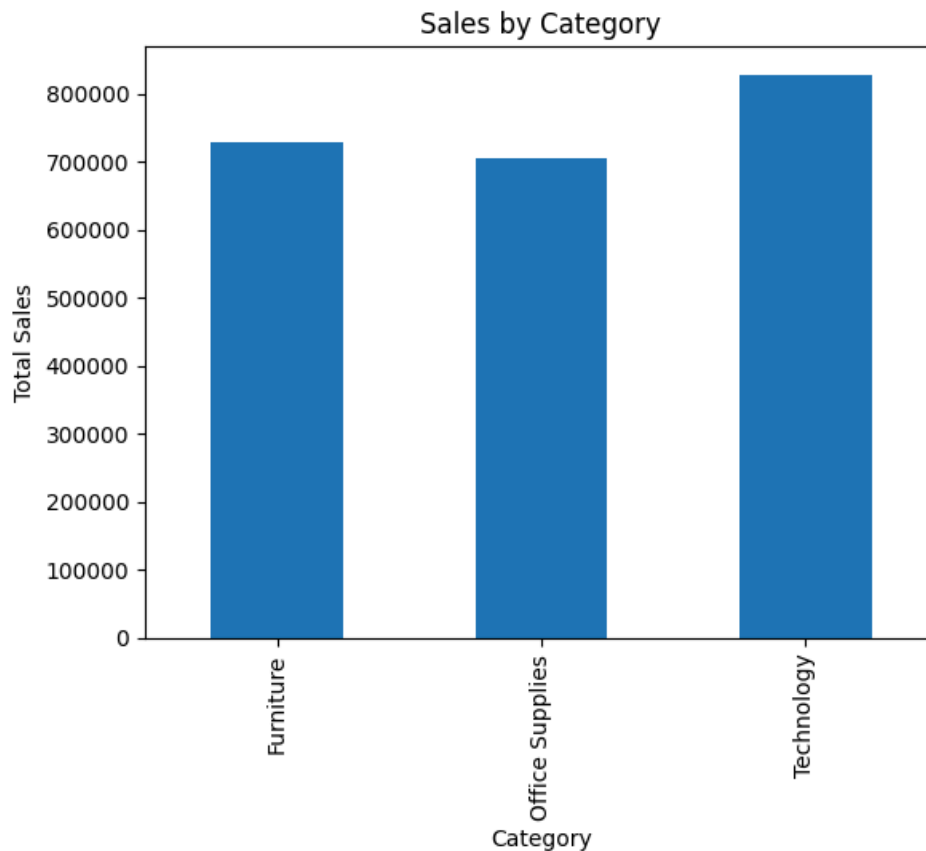
# Average Order Value
avg_order_value = df.groupby('Order_ID')['Sales'].sum().mean()

print("Total Sales:", total_sales)
print("Total Orders:", total_orders)
print("Average Order Value:", avg_order_value)
```

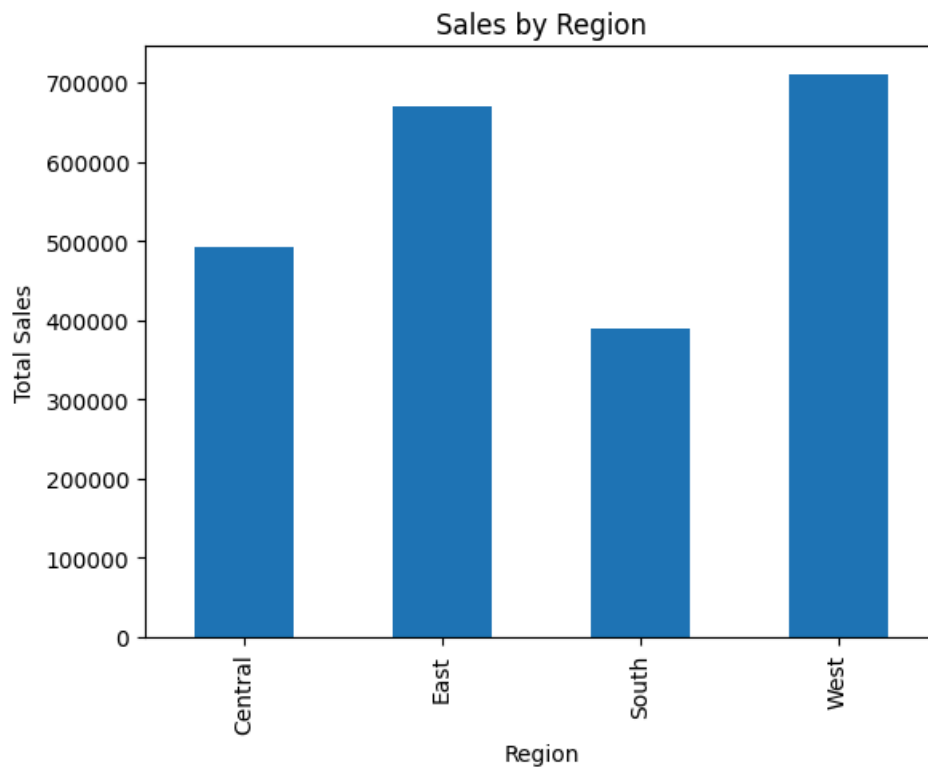
```
Total Sales: 2261536.7827000003
Total Orders: 4922
Average Order Value: 459.4751691791955
```

```
import matplotlib.pyplot as plt

df.groupby('Category')['Sales'].sum().plot(kind='bar')
plt.title('Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.show()
```



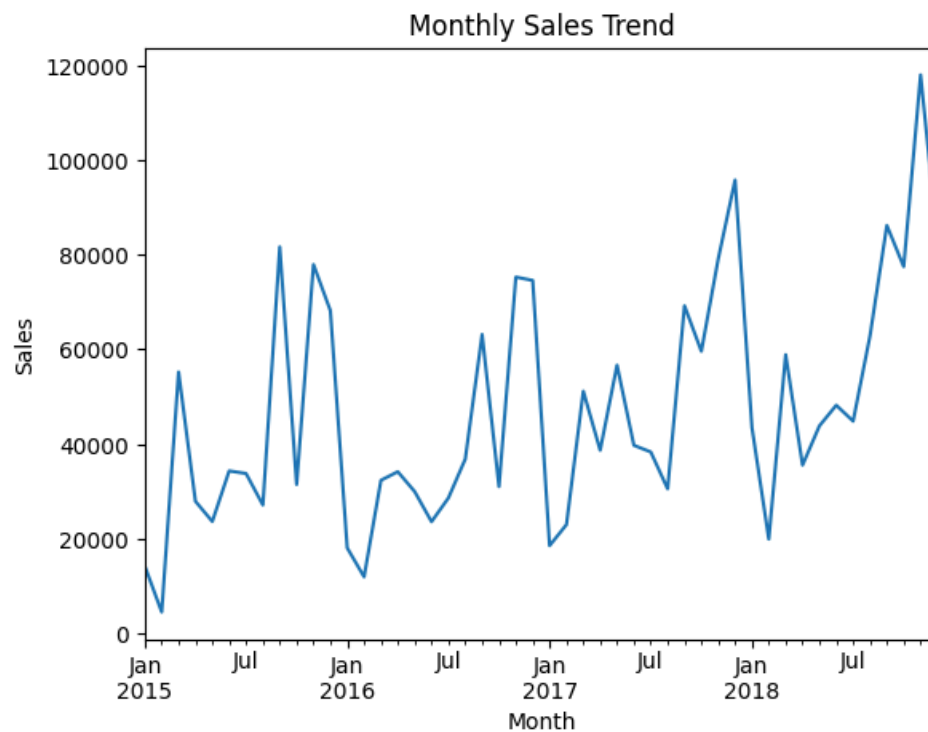
```
df.groupby('Region')['Sales'].sum().plot(kind='bar')
plt.title('Sales by Region')
plt.xlabel('Region')
plt.ylabel('Total Sales')
plt.show()
```



```
monthly_sales = df.resample('M', on='Order_Date')['Sales'].sum()
```

```
monthly_sales.plot()  
plt.title('Monthly Sales Trend')  
plt.xlabel('Month')  
plt.ylabel('Sales')  
plt.show()
```

/tmp/ipython-input-2584954596.py:1: FutureWarning: 'M' is deprecated and will be removed in a future
monthly_sales = df.resample('M', on='Order_Date')['Sales'].sum()



```
df.groupby('Product_Name')['Sales'].sum().sort_values(ascending=False).head(10)
```

	Sales
Product_Name	
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24 Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

dtype: float64